Special Topics: Social Media Mining and Analysis – Assignment 1

Unimodal Baselines for Hateful Meme Classification

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https://github.com/BogdanFanea/SpecialTopicSocialMining

We tested four pretrained encoders on the Hateful Memes dataset [1], using either images or the supplied meme text. Each encoder stays frozen while a separate linear classifier learns from its features. All four classifiers use the same design.

1. Data and evaluation labels

Split	Examples	Non-hateful (0)	Hateful (1)
Training	8,500	5,450	3,050
Validation (`dev`)	500	250	250
Test (public-mirror labels)	1,000	510	490

The splits have no shared IDs, and all images are present. We kept the supplied training and validation labels. The course test file has no labels, so we used a pinned public mirror [5] with the same 1,000 IDs, texts, and image paths. Its test annotation file is byte-identical to the Kaggle copy cited by a research implementation. **Official ground-truth provenance is unconfirmed.** Label versions differ: the mirror changes 823 training and three validation labels; we apply neither change. Test scores below are therefore *AUROC against public-mirror labels*, not authenticated competition scores. Source revision and checksums are recorded in the accompanying provenance file.

2. Encoders and preprocessing

Baseline	Exact pretrained checkpoint	Dim.	Pooling
Image A	`ResNet50_Weights.IMAGENET1K_V2`	2,048	Global average
Image B	`openai/clip-vit-base-patch32`	512	Vision CLS + projection
Text A	`sentence-transformers/all-MiniLM-L6-v2`	384	Masked mean + L2
Text B	`sentence-transformers/all-mpnet-base-v2`	768	Masked mean + L2

We chose ResNet-50 as an ImageNet-trained CNN baseline and CLIP as a vision transformer trained with image–text pairs [2, 3]. MiniLM is a compact six-layer sentence encoder; MPNet is a larger twelve-layer model [4]. Because these encoders differ in architecture, size, and pretraining, the results cannot isolate the effect of any one of these factors.

Image branches receive **RGB pixels only**. ResNet uses its official bilinear shorter-edge resize to 232 and 224×224 center crop with ImageNet channel normalization. CLIP uses the checkpoint's bicubic resize and 224×224 center crop with CLIP channel normalization; its text tower is unused. Text branches receive **only the provided meme text**, using their matching tokenizers, dynamic padding, and truncation limits of 256 (MiniLM) and 384 (MPNet) tokens, including special tokens. Attention-mask mean pooling excludes padding, includes non-padding special tokens, and is followed by L2 normalization. No text was truncated in any split. Encoders run in evaluation/inference mode; float32 embeddings are cached once. Exact model revisions and normalization constants are documented in `config.json` and the README.

3. Linear probes and model selection

For every encoder we fit $p(y = 1 \mid z) = \sigma(w^\top \widetilde{z} + b)$, where $\widetilde{z}$ uses per-dimension mean/scale fitted **only on training embeddings**. The classifier is `Linear(d,1)` with bias and mean `BCEWithLogitsLoss`; only d varies. AdamW uses batch size 256, 50 epochs, learning rates $\{10^{-4}, 10^{-3}, 10^{-2}\}$ and weight decay $\{0, 0.01\}$ (including bias), with default betas/epsilon and no scheduler or class weighting. We ran all six configurations with seeds $\{42, 43, 44\}$: 72 fits in total. For each fit, we kept the epoch with the highest validation AUROC, then chose settings by mean validation AUROC across seeds. Ties keep the earliest epoch or first grid entry. We evaluated on test after selection; validation and test data were never used for training.

Selected settings (learning rate, weight decay; best epochs for seeds 42/43/44): ResNet $(.001, .01;\ 1/2/1)$; CLIP $(.01, .01;\ 4/10/9)$; MiniLM $(.001, 0;\ 5/8/10)$; MPNet $(.001, .01;\ 5/2/1)$. Extraction used Apple M3 MPS (16 GB); deterministic, single-thread CPU probe training used PyTorch 2.4.0 and Transformers 4.44.2.

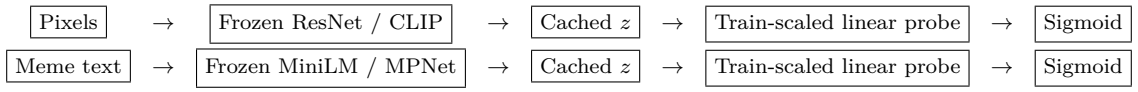


Figure 1. Four independent branches; the alternatives in each row are run separately.

4. Results and comparison

Table 1 reports AUROC on the 0–1 scale, computed from continuous sigmoid hateful-class scores. Values are mean ± sample standard deviation over the same three seeds; predictions are not ensembled.

Baseline	Encoder checkpoint	Validation AUROC	Test AUROC†
Image A	`ResNet50_Weights.IMAGENET1K_V2`	0.5429 ± 0.0218	0.5390 ± 0.0091
Image B	`openai/clip-vit-base-patch32`	$\mathbf{0.6445} \pm 0.0049$	0.6552 ± 0.0033
Text A	`sentence-transformers/all-MiniLM-L6-v2`	0.6310 ± 0.0019	$\mathbf{0.6603} \pm 0.0016$
Text B	`sentence-transformers/all-mpnet-base-v2`	0.6390 ± 0.0031	0.6566 ± 0.0069

Table 1. †AUROC against public-mirror labels; see Section 1. Bold marks the highest mean in each column.

CLIP scored higher than ResNet-50 on both splits. MPNet was the stronger text model on validation, but MiniLM had the higher test mean, so the larger text encoder did not perform better on both splits. We selected CLIP and MPNet for the case analysis using validation scores. Their test means, 0.6552 and 0.6566, differ by only 0.0014. Even MiniLM's highest observed test mean exceeds CLIP by just 0.0051. These small differences do not establish a reliable modality advantage: three seeds measure optimizer variability, not sampling confidence, and validation estimates were reused for selection.

CLIP's training on image–text pairs may help it capture meaning beyond ImageNet object categories, though this comparison cannot establish the cause. Text models can use words associated with hate while missing what an image refers to or whether a caption rejects a hateful message. Confounder pairs challenge both input types. **Image models still receive the written captions as pixels**, and CLIP's pretraining used text. Center cropping may also remove useful context at image edges.

5. Five validation disagreements

We compare CLIP and MPNet using seed 42 and the fixed rule **hateful iff score** ≥ 0.5. They disagree on 167 of 500 validation examples. We inspect the five disagreements with the largest absolute score gap (ties by ascending ID). In all five cases, CLIP predicts non-hateful and MPNet predicts hateful. These selected examples show possible errors, but not how common each error is. AUROC itself uses no threshold. Labels below are the supplied course labels.

ID	True y	p_{image}	p_{text}	Inspected context and likely error
38047	0	0.160	0.853	A gendered insult overlays a political group photo. MPNet flags offensive wording; the supplied non-hateful label is ambiguous, highlighting the distinction between offensiveness and group targeting.
82509	0	0.145	0.818	A runway photo accompanies a vulgar "72 virgins" caption. MPNet may overreact to provocative wording without resolving the target; the supplied label is non-hateful.
01967	0	0.118	0.765	A burning swastika resolves the caption's reference to racism's "symbol." MPNet's false positive may miss the anti-racist stance and the visual referent.
48296	1	0.156	0.771	A woman playfully bites a dog's ear; a caption about an Asian boyfriend's family invokes a racist dog-eating stereotype. CLIP misses this image–text connection.
19536	1	0.052	0.664	An altered-nose portrait and a caption about a Jewish friend finding coins invoke antisemitic stereotypes. CLIP misses the combination; MPNet can exploit explicit lexical cues.

These explanations come from inspecting the images and captions; we did not test which features drove each decision. A correct text prediction does not show that a model understood a joke that depends on the image. Ambiguous labels and the uncertain source of the test labels also limit the conclusions. We did not change any labels.

Sources and reused implementations

The code uses TorchVision/Hugging Face model loaders, PyTorch optimization, and scikit-learn scaling/metrics. Masked pooling follows the text model-card examples. Configurations, all selected classifier checkpoints, per-seed predictions, and complete search histories accompany the submission.

References

[1] D. Kiela et al. The Hateful Memes Challenge: Detecting Hate Speech in Multimodal Memes. NeurIPS, 2020.
[2] TorchVision. ResNet-50 pretrained weights and inference transforms, version 0.19.
[3] A. Radford et al. Learning Transferable Visual Models From Natural Language Supervision. ICML, 2021. CLIP checkpoint.
[4] Sentence Transformers. Model cards and pooling examples for all-MiniLM-L6-v2 and all-mpnet-base-v2.
[5] Public test-label mirror, revision d201c488; Missing-Aware Prompts data instructions. Accessed 17 October 2026.